



E-COMMERCE WEBSITE FOR A LOCAL ARTISAN COOPERATIVE

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Contents

Introduction.....	3
Scenario	3
Literature Review	5
SDLC: DevOps	6
Stakeholders	8
Functional/Non-functional Requirements.....	9
Mapping requirements	14
Context Diagram.....	16
Use Case Diagram	18
Class Diagram	20
Interface Design.....	23
Architecture of The System using MVC.....	25
Conclusion	26

Introduction

The rapid growth of digital platforms has significantly transformed the way individuals and businesses interact, trade, and collaborate. In particular, e-commerce systems have become essential tools for connecting sellers with a wider audience while offering customers convenient, secure, and efficient purchasing experiences. Alongside traditional online stores, auction-based marketplaces have gained increasing popularity due to their ability to create competitive pricing environments and enhance user engagement. These technologies are especially valuable for small-scale producers and artisan cooperatives seeking broader market reach and sustainable business opportunities.

This project presents the design and analysis of an **E-Commerce Website for a Local Artisan Cooperative**, developed to support both fixed-price product sales and real-time online auctions. The system aims to provide a unified digital marketplace where customers can browse products, place orders, participate in auctions, and track their purchases, while artisans can manage their shops, products, orders, and sales performance through dedicated dashboards. The platform is designed to ensure security, usability, scalability, and compliance with modern e-commerce standards, making it suitable for both local and international users.

The report follows a **Software Engineering** approach, applying the **DevOps-based Software Development Life Cycle (SDLC)** to emphasize continuous integration, continuous deployment, and ongoing feedback. Security and compliance considerations are treated as core concerns throughout every development phase, reflecting best practices in modern system design. Functional, non-functional, business, and regulatory requirements are systematically identified and mapped to ensure that the system satisfies stakeholder needs while maintaining high performance and reliability.

By combining e-commerce functionalities, auction mechanisms, and cooperative values within a single platform, this project demonstrates how software engineering principles can be applied to build a robust, user-centric, and secure digital marketplace. The report documents the system's requirements, architecture, diagrams, and design decisions, providing a comprehensive blueprint for the development of a real-world cooperative online marketplace.

Scenario

When a new customer visits the cooperative's website, they can register by entering their full name, a valid email address, and a secure password. The system ensures that the provided email follows the correct format before completing registration, and all passwords are safely encrypted before being stored. Artisans, on the other hand, can create accounts by not only submitting their email and password but also by providing shop details such as their biography, profile picture, and shop name. Once registered, both customers and artisans can log in to the system using their credentials. For security and flexibility, the platform also offers logout functionality and password recovery through email verification in case users forget their login details.

After logging in, customers are welcomed with access to the product catalogue, where they can browse and search for products using keywords, artisan names, or product categories like painting, jewellery, pottery, or textiles. Each product page displays clear details, including the product name, description, price, multiple images, and artisan profile information. Customers can add items they like to their shopping cart, review their cart, make adjustments such as changing item quantities or removing products, and instantly see the updated total cost. Once ready to make a purchase, they follow a guided checkout process where they provide a shipping address and payment details, after which the system generates an order confirmation and sends a confirmation email to the customer. From their user account, customers can also track the status of their orders — from pending to shipped and finally delivered.

Artisans experience the platform through a dedicated dashboard that allows them to manage their profiles and products effectively. They can add new products with descriptions and images, as well as edit or remove existing listings. When a customer places an order, the artisan receives a notification and is able to update the order's status, ensuring customers are aware of shipping progress. Alongside product management, artisans can also review their complete sales history and examine performance reports to understand their business growth.

The system manages all orders with unique identifiers, enabling both customers and artisans to view detailed order histories. While relevant transaction details are stored to maintain accurate records, sensitive payment data is never kept within the platform. Payments are processed securely through integrated payment gateways, and customers receive email confirmations once payments are completed successfully.

For a more dynamic marketplace, artisans also have the option to list their products in auctions. To create an auction, artisans provide details such as the product description, a starting bid, and the auction's start and end times. Customers can then place bids, with the system enforcing that any new bid must be higher than the current highest one. The auction interface displays the leading bid and bidder details in real time, and customers receive notifications if they are outbid. At the scheduled end time, the auction closes automatically, and both the winning bidder and artisan are notified. The winning customer is then guided to complete the payment for the item. The marketplace separates fixed-price products from active and upcoming auctions for easier browsing.

To enhance user experience, the cooperative's website includes extra features such as product and artisan reviews, where customers can leave ratings and feedback. They can also create and maintain personalized wishlists. For international reach, the platform supports multiple languages and currencies, making accessibility easier for customers worldwide. Finally, administrators have access to a dedicated dashboard from which they can manage all platform activities, including users, artisans, products, orders, and auctions, ensuring smooth and secure operations for the entire cooperative.

Literature Review

The literature review highlights several platforms that closely reflect the concept of an online auction website for artworks, with each system offering a distinct model that can be leveraged.

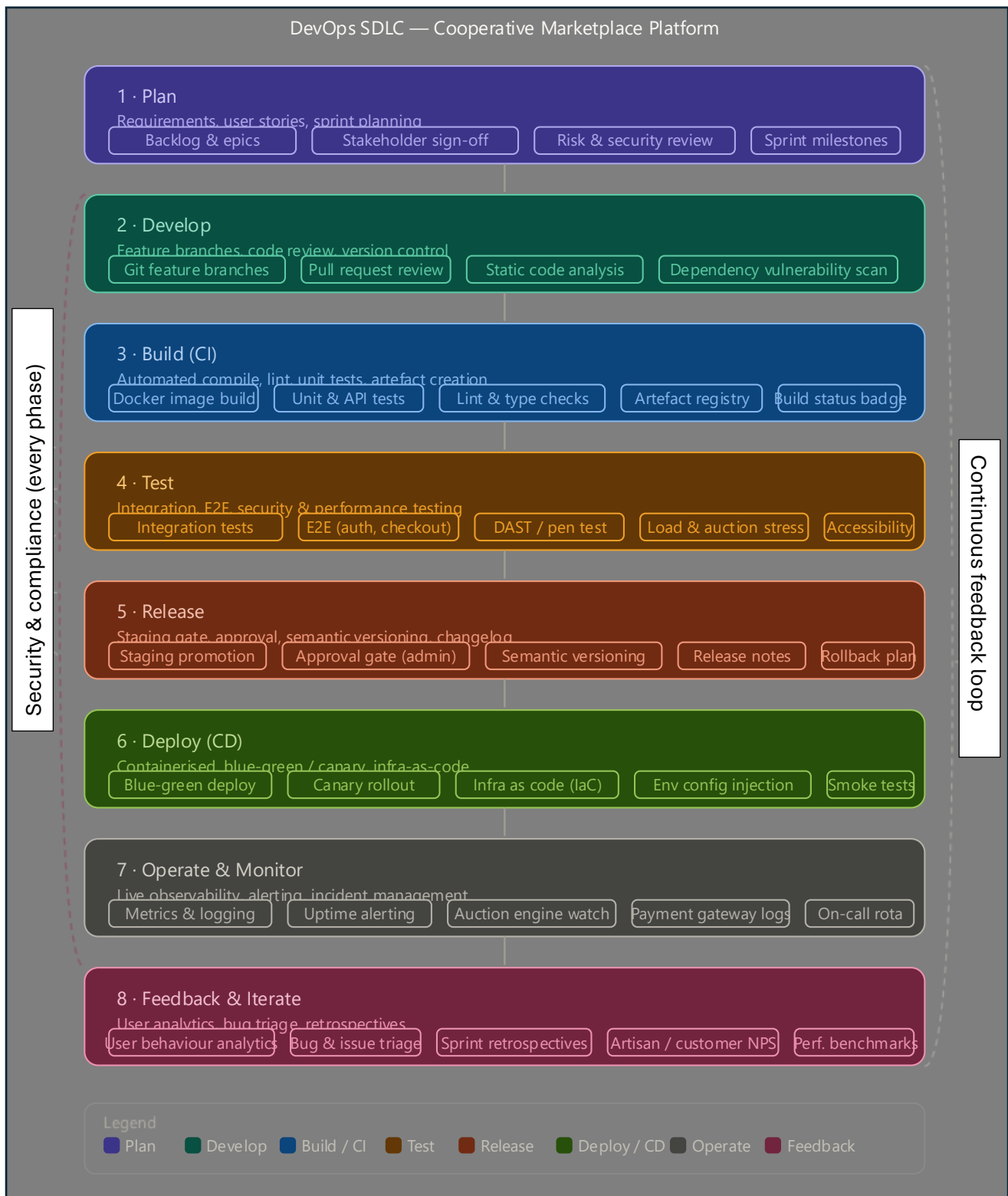
eBay is one of the most well-known platforms that relies on an open bidding system. It allows users to list items and run auctions where buyers from around the world can place bids. The platform is highly flexible and user-friendly, making it suitable for a wide variety of products, including artworks. It also provides useful features such as setting a starting price, defining auction duration, and offering a “Buy It Now” option, giving sellers greater control over how they present their items and maximize sales opportunities.

In contrast, Catawiki operates as a curated auction platform where listings are reviewed by experts before being published. It focuses on unique items such as artworks and rare collectibles, ensuring quality and authenticity. This approach builds trust among users and attracts buyers who are specifically looking for distinctive and verified pieces, making it a strong model for specialized art auction platforms.

On the other hand, **Sotheby’s** represents a high-end and professional model in the art auction industry. As a historic auction house, it combines traditional auction practices with modern digital access, allowing users to participate in auctions both online and in person. The platform focuses on premium, high-value artworks and offers advanced features such as live-streamed auctions and professional art evaluations.

Together, these platforms demonstrate how auction systems can range from simple and accessible to highly specialized and professional, while sharing the common goal of creating competitive bidding environments, increasing the value of artworks, and connecting sellers with a global audience.

SDLC: DevOps



The diagram presents the Software Development Life Cycle (SDLC) for the cooperative marketplace platform using a DevOps model, illustrated as a linear pipeline of eight sequential phases with a continuous feedback loop connecting the final phase back to the first. A

dedicated security and compliance track runs alongside every phase, reinforcing that security is not a single checkpoint but an ongoing obligation throughout the entire development process.

The first phase, **Plan**, captures all requirements engineering activities, including backlog creation, epic definition, stakeholder sign-off, risk and security review, and sprint milestone planning. This phase lays the foundation for the platform's domain areas, covering user registration and authentication, product management, auction mechanics, order processing, and the administrator dashboard.

The second phase, **Develop**, represents the coding and collaboration workflow. Developers work on isolated Git feature branches, submit pull requests for peer review, and run static code analysis and dependency vulnerability scans before any code is merged. This is particularly important given the multi-role nature of the system, which includes customers, artisans, and administrators with distinct access levels and permissions.

The third phase, **Build (CI)**, is the Continuous Integration stage, where every code merge triggers an automated pipeline that compiles the application, runs linting and type checks, executes unit and API tests, and produces a versioned Docker image stored in an artefact registry. A build status badge provides immediate visibility into pipeline health.

The fourth phase, **Test**, encompasses a comprehensive quality assurance process. It includes integration tests, end-to-end tests covering critical user journeys such as authentication and checkout, dynamic application security testing (DAST) and penetration testing, load and auction stress tests to validate the real-time bidding engine under high concurrency, and accessibility testing to support the platform's multilingual and international audience.

The fifth phase, **Release**, acts as a formal gate before any code reaches production. It involves promotion to a staging environment, an administrative approval gate, semantic versioning, the generation of release notes and changelogs, and the preparation of a rollback plan in case a deployment needs to be reversed.

The sixth phase, **Deploy (CD)**, is the Continuous Delivery stage. The platform is deployed using blue-green and canary release strategies to minimise downtime — a critical consideration for a live auction system where interruptions would directly impact active bids. Infrastructure as Code (IaC) manages environment provisioning, environment-specific configuration is injected at runtime, and automated smoke tests verify the deployment immediately after it completes.

The seventh phase, **Operate and Monitor**, covers the live observability of the running system. It includes centralised metrics collection and logging, uptime alerting, dedicated monitoring of the auction engine, payment gateway log auditing, and an on-call rotation to ensure rapid incident response.

The eighth and final phase, **Feedback and Iterate**, closes the DevOps loop. User behaviour analytics, bug and issue triage, sprint retrospectives, artisan and customer satisfaction scores (NPS), and performance benchmarks are all collected and fed back into the planning phase, driving continuous improvement across the platform.

The dashed loop on the right side of the diagram represents this continuous feedback cycle, visually reinforcing that DevOps is not a linear process that ends at deployment, but an iterative, self-improving system. The dashed track on the left, labelled "Security and compliance (every phase)", underscores the platform's obligation to protect sensitive data — including encrypted passwords, personally identifiable information, and payment transaction records — throughout every stage of development and operations.

Stakeholders

- 1. Customers** Registered users who browse the product catalogue, place orders, participate in auctions, make payments, submit reviews, manage wishlists, and track order delivery status through their accounts.
- 2. Artisans** Registered sellers who create and manage shop profiles, list fixed-price and auction products, receive and update order statuses, view sales history, and monitor performance reports through their dedicated dashboard.
- 3. System Administrators** Internal operators who oversee the entire platform through the admin dashboard, managing users, artisans, products, orders, and auctions to ensure smooth, secure, and compliant platform operations.
- 4. Authentication System** A technical stakeholder responsible for validating user credentials, enforcing email format rules, encrypting and storing passwords securely, managing session tokens, and supporting password recovery flows via email verification.
- 5. Email Service Provider** An external service that handles all outgoing platform emails, including registration confirmations, order confirmations, payment receipts, password recovery links, auction outbid notifications, and auction win notifications.
- 6. Payment Gateway** An external financial service that securely processes customer payments at checkout and after auction wins, ensuring that no sensitive payment data is stored within the platform itself.
- 7. Database System** A technical stakeholder responsible for persisting all platform data, including user profiles, product listings, order records, auction bids, reviews, wishlists, and transaction histories, while enforcing data integrity and access control.
- 8. Shipping Providers** External logistics partners who handle the physical delivery of purchased products, supplying tracking information that feeds into the order status updates visible to both customers and artisans.
- 9. Regulatory Authorities** External governmental and legal bodies whose data protection, consumer rights, financial compliance, and e-commerce regulations the platform must adhere to, particularly given its international and multi-currency reach.
- 10. Translation / Localisation Services** External or integrated services responsible for translating platform content and adapting currency displays to support the multilingual and multi-currency requirements of the platform's international customer base.

11. Non-registered Visitors Unauthenticated users who can browse the public-facing areas of the platform, such as the product catalogue and artisan profiles, but who cannot place orders, bid in auctions, or access account-specific features without registering.

12. Developers / Maintenance Team Internal technical staff responsible for building, testing, deploying, and maintaining the platform, resolving bugs, releasing updates, monitoring system health, and ensuring the reliability and security of all platform components over time.

Functional/Non-functional Requirements

1. Functional Requirements (FR)

- **FR1 (User Registration):** The system shall provide two distinct sign-up forms: one for Customers (Name, Email, Password) and one for Artisans (Email, Password, Shop Name, 500-character Bio, and Profile Image upload).
- **FR2 (Email Validation):** The registration module must verify that email inputs contain an "@" symbol and a valid domain extension before allowing form submission.
- **FR3 (Authentication & Recovery):** The system shall implement session-based login and a password reset feature that sends a unique, 15-minute expiring link to the user's verified email.
- **FR4 (Product Browsing):** The catalog shall support keyword search and filtering by four predefined categories: Painting, Jewellery, Pottery, and Textiles.
- **FR5 (Cart Management):** The cart shall calculate a running total of all items, including quantity adjustments, and persist data for logged-in users for up to 30 days.
- **FR6 (Order Processing):** Upon checkout completion, the system shall generate a 10-digit unique Order ID and trigger an automated confirmation email within 60 seconds.

2. Non-Functional Requirements (NFR)

- **NFR1 (Usability):** New users should be able to complete a purchase from the homepage in 5 clicks or fewer without external assistance.
- **NFR2 (Availability):** The platform shall maintain a 99.9% uptime, allowing no more than 43 minutes of unplanned downtime per month.
- **NFR3 (Performance):** All pages must achieve a Largest Contentful Paint (LCP) of under 2.5 seconds on a standard 4G connection.
- **NFR4 (Scalability):** The system must handle a 300% spike in traffic (up to 1,000 concurrent users) during holiday sales without a drop in response time.
- **NFR5 (Data Security):** All user data at rest must be encrypted using AES-256 standards.

3. User Requirements (UR)

- **UR1 (Purchase Journey):** Customers require a "One-Click" add-to-cart feature directly from search results to streamline the buying process.
- **UR2 (Artisan Management):** Artisans require a visual "Sales Performance" chart on their dashboard showing monthly revenue trends.
- **UR3 (Secure Access):** Users require visual confirmation (e.g., a lock icon) and Multi-Factor Authentication (MFA) options for high-value transactions.
- **UR4 (Order Tracking):** Customers require a real-time status bar (Pending > Shipped > Delivered) accessible via their "My Orders" page.
- **UR5 (Admin Control):** Administrators require the ability to "Shadow" or "Suspend" accounts that violate the cooperative's terms of service within two clicks.

4. System Requirements (SR)

- **SR1 (Web Accessibility):** The system must be accessible via any modern browser (Chrome, Firefox, Safari, Edge) and support screen readers (WCAG 2.1 Level AA compliance).
- **SR2 (Payment Integration):** The system shall integrate with external APIs (Stripe/PayPal) to process Visa, Mastercard, and Amex payments.
- **SR3 (Data Storage):** The system shall utilize a relational database (e.g., PostgreSQL) to maintain ACID-compliant records of all transactions.
- **SR4 (Notification Engine):** The system shall integrate with an SMTP server to deliver 100% of system-generated alerts to users' inboxes.
- **SR5 (Concurrency):** The backend must support at least 200 simultaneous database writes per second during high-volume auction bidding.

5. Business Requirements (BR)

- **BR1 (Artisan Promotion):** The homepage must reserve at least 40% of its "Above the Fold" space for rotating "Artisan of the Week" features.
- **BR2 (Marketplace Trust):** The platform shall implement a "Verified Artisan" badge for sellers who have successfully completed 50+ orders with a 4.5-star rating.
- **BR3 (Revenue Growth):** The system must support a 10% commission fee automatically deducted from each non-auction sale to fund cooperative operations.
- **BR4 (Global Reach):** The platform shall implement SEO-friendly URLs for all products to increase organic search visibility by 20% annually.
- **BR5 (Transaction Efficiency):** Total checkout time from "Cart" to "Confirmation" should average less than 90 seconds for returning users.

6. Interface Requirements (IR)

- **IR1 (Web UI):** The interface shall use a consistent color palette and typography (Design System) across both Customer and Artisan views.
- **IR2 (Dashboards):** The Artisan Dashboard must provide a distinct "Inventory Low" alert when product stock falls below 3 units.
- **IR3 (Mobile Design):** The UI must be fully responsive, with all touch targets (buttons) being at least 44x44 pixels on mobile devices.
- **IR4 (Payment UI):** The payment interface must be hosted via a secure, PCI-compliant iframe to ensure no sensitive card data is handled by the local UI.
- **IR5 (Email UI):** All system emails must be formatted in responsive HTML/CSS to ensure readability on both mobile and desktop mail clients.

7. Performance Requirements (PR)

- **PR1 (Multi-User Support):** The system must support up to 5,000 registered users with 500 active sessions simultaneously.
- **PR2 (Transaction Speed):** Payment gateway handshakes must be completed within 5 seconds for 99% of all transactions.
- **PR3 (Search Latency):** Search results for a database of 10,000 products must be returned in under 800ms.
- **PR4 (Auction Real-Time):** Bid updates must be pushed to all active viewers via WebSockets within 300ms of the bid being placed.
- **PR5 (Peak Load):** During the final 5 minutes of an auction, the system shall prioritize bid processing over non-essential page requests.

8. Security Requirements (SEC)

- **SEC1 (Encryption):** All passwords must be hashed using the Argon2 or BCrypt algorithm with a unique salt per user.
- **SEC2 (Authorization):** The system shall use Role-Based Access Control (RBAC) to ensure Artisans cannot access Admin dashboards or other Artisans' sales data.
- **SEC3 (Zero-Storage Policy):** The platform must not store CVV codes or full Credit Card Numbers in its local database or logs.
- **SEC4 (Brute Force Protection):** The system shall lock any account for 30 minutes after 5 consecutive failed login attempts.
- **SEC5 (HTTPS):** All data transmission must be forced over TLS 1.3 (HTTPS) with an A+ rating on SSL Labs.

9. Regulatory / Compliance Requirements (CR)

- **CR1 (GDPR/Data Laws):** The system must include a "Cookie Consent" banner and a "Right to be Forgotten" button within the user settings.
- **CR2 (Privacy):** A clearly linked Privacy Policy must be accessible from every page footer, detailing how user data is utilized.
- **CR3 (PCI-DSS):** The system must comply with PCI-DSS Level 4 requirements for secure online payment processing.
- **CR4 (E-Commerce Standards):** The system shall provide automated digital tax invoices for every purchase to comply with local trade regulations.
- **CR5 (Audit Trails):** All financial transactions must be logged in a read-only audit file accessible only by the System Administrator.

10. Operational Requirements (OR)

- **OR1 (Maintenance):** Scheduled maintenance windows must be communicated to users 24 hours in advance and occur during low-traffic hours (2 AM – 4 AM).
- **OR2 (Backup):** The system shall perform full database backups every 24 hours and transaction log backups every 15 minutes to an off-site cloud bucket.
- **OR3 (Monitoring):** The system shall use an automated monitoring tool (e.g., Datadog/New Relic) to alert admins via SMS if CPU usage exceeds 90%.
- **OR4 (Support Logs):** The system must maintain a log of all customer support tickets and their resolution times for quarterly review.
- **OR5 (Recovery):** The Disaster Recovery Plan must guarantee a Recovery Point Objective (RPO) of 15 minutes.

11. Domain Requirements (DR)

- **DR1 (Categorization):** Every product uploaded by an artisan must be assigned to at least one of the four domain-specific categories (Painting, Jewellery, Pottery, Textiles).
- **DR2 (Auction Logic):** Auctions must have a mandatory "Start Price" and a "End Timestamp" set at the time of creation.
- **DR3 (Bid Validation):** The system shall reject any bid that does not exceed the current highest bid by at least the "Minimum Increment" defined for that auction.
- **DR4 (Order IDs):** Order identifiers must follow a standard prefix (e.g., COOP-2024-XXXXX) for easy identification by the cooperative.
- **DR5 (Multi-Currency):** The system shall update currency exchange rates via a third-party API every 12 hours.

12. Inverse Requirements (IRV)

- **IRV1 (Payment Storage):** The system shall **not** allow the local database to save raw credit card data under any circumstances.
- **IRV2 (Invalid Email):** The system shall **not** proceed with registration if the email validation check returns a false or "disposable email" status.
- **IRV3 (Illegal Bidding):** The system shall **not** accept a bid from the artisan who owns the product being auctioned (shill bidding prevention).
- **IRV4 (Cross-Account Access):** The system shall **not** allow a Customer to view the "Artisan Dashboard" unless they have upgraded their account type.
- **IRV5 (Data Deletion):** The system shall **not** allow the deletion of an order record if it is currently in "Shipped" or "Pending" status.

13. Design Requirements (DRQ)

- **DRQ1 (UI/UX):** The design shall use "High Contrast" settings (contrast ratio 4.5:1) for all text to assist visually impaired users.
- **DRQ2 (Architecture):** The system shall use a Microservices or Modular Monolith architecture to allow the "Auction" module to be updated without taking down the "Storefront."
- **DRQ3 (Scalable Assets):** All images must be served in WebP format and automatically resized based on the user's screen resolution.
- **DRQ4 (Navigation):** The "Breadcrumb" navigation style must be present on all product and auction pages to orient the user.
- **DRQ5 (Layout):** The system must maintain a "Sticky Header" on desktop views to keep the Search bar and Cart icon visible at all times.

14. Implementation Requirements (IRQ)

- **IRQ1 (Coding Standards):** All code must pass a static analysis tool (e.g., SonarQube) with zero "Critical" or "Blocker" vulnerabilities before deployment.
- **IRQ2 (Database):** The database implementation must include "Foreign Key" constraints to prevent orphaned order records.
- **IRQ3 (API Integration):** External API calls must have a 10-second timeout to prevent the system from hanging if a third-party service is slow.
- **IRQ4 (Data Encryption):** Personal Identifiable Information (PII) such as phone numbers and addresses must be encrypted at the database column level.
- **IRQ5 (Testing):** The system must undergo "Load Testing" to simulate 10,000 users and "Penetration Testing" to identify security flaws before the official launch.

Mapping requirements

i. The Artisan Management

- **Profile Creation:** The system shall allow artisans to create a digital storefront by submitting an email, password, shop name, and a text-based biography of up to 1,000 characters.
 - **Product Lifecycle:** Artisans must be able to add, edit, and delete product listings, including a minimum of three high-resolution images and a numerical price field.
 - **Auction Configuration:** The system shall provide a dedicated interface for artisans to set "Starting Bids" and "End Timestamps" for individual product auctions.
 - **Order Fulfillment:** Artisans shall have the functionality to update an order's status from "Pending" to "Shipped" and "Delivered" via a one-click toggle in their dashboard.
-

ii. Shopping Management

- **Catalog Search & Filter:** Customers must be able to filter search results by four categories (Painting, Jewellery, Pottery, Textiles) with a response time of less than 800ms for 10,000 items.
 - **Cart Persistence:** The shopping cart must retain all added items and their specific quantities for up to 30 days for logged-in users, recalculating the total cost instantly upon any change.
 - **Real-Time Bidding:** The auction system must refresh the "Current Highest Bid" and "Leading Bidder" details for all active viewers every 500ms using WebSocket technology.
 - **Wishlist & Reviews:** Customers shall be able to save up to 50 items to a personal wishlist and submit text-based reviews with a 1-to-5-star numerical rating.
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iii. Payment Process

- **Third-Party Integration:** The system shall hand off all transaction data to an external, PCI-compliant payment gateway (e.g., Stripe) without storing the customer's full credit card number locally.
- **Currency Conversion:** The platform must support real-time currency conversion based on exchange rates updated every 12 hours via an external financial API.
- **Automated Invoicing:** Within 60 seconds of a successful transaction, the system must generate a PDF tax invoice and deliver it to the customer's verified email address.

- **Payment Verification:** The system must reject any transaction where the billing address does not match the payment method's authorization records.
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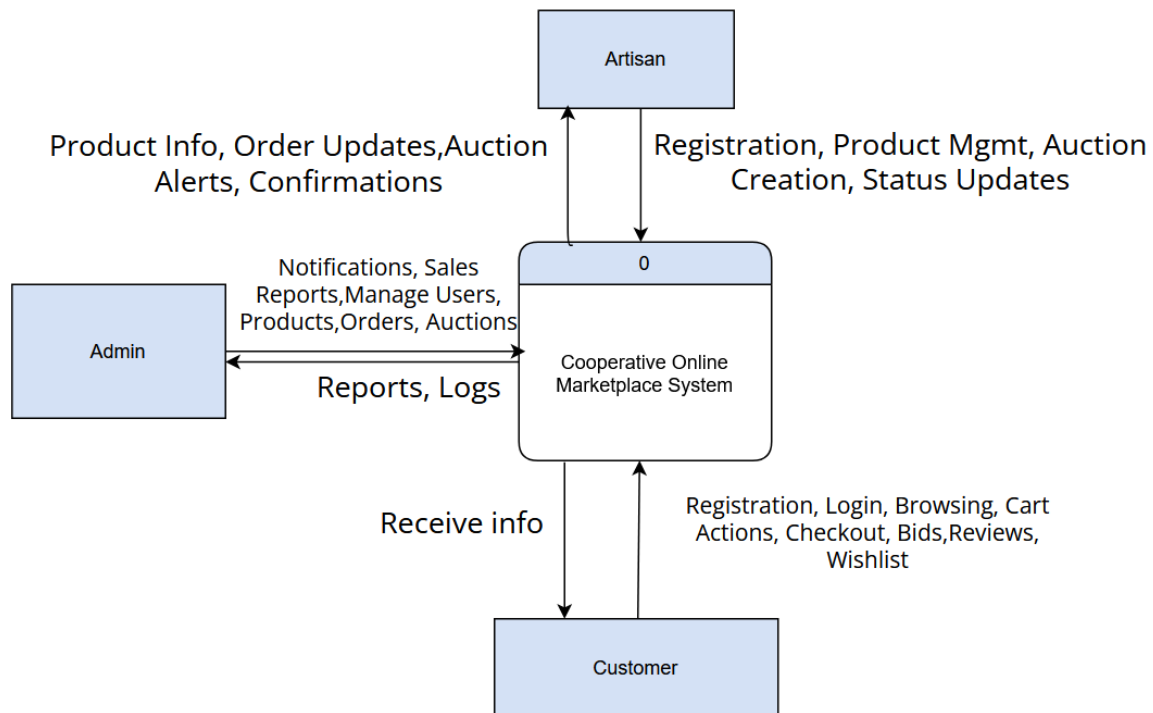
iv. Reporting Management

- **Sales History:** The system shall provide artisans with a searchable table of all past sales, including date, item name, buyer ID, and final transaction amount.
 - **Performance Dashboards:** Artisans must have access to a visual line chart representing their monthly revenue and a bar chart showing their "Most Viewed" products.
 - **Admin Logs:** The system shall generate a daily report for administrators summarizing total marketplace revenue, total new registrations, and active auction counts.
 - **Auction Summaries:** Upon the closing of an auction, the system must automatically generate a "Winning Bid Report" sent to both the artisan and the successful bidder.
-

v. Security & System Administration (Added Functions)

- **Role-Based Access Control (RBAC):** The system must restrict access so that Artisans cannot view Admin dashboards and Customers cannot access Artisan-specific management tools.
- **Account Protection:** The system shall automatically lock any user account for 30 minutes following 5 consecutive failed login attempts.
- **Administrative Oversight:** Administrators must be able to flag, suspend, or delete any product listing or user account that violates cooperative policies within a single "Manage" interface.
- **Data Integrity & Recovery:** The system shall perform incremental database backups every 6 hours with a guaranteed Recovery Time Objective (RTO) of under 2 hours in case of system failure.

Context Diagram



This diagram is a **Context Diagram (Level 0 Data Flow Diagram)** for the **Cooperative Online Marketplace System**. It illustrates the high-level interactions between the core system and its external entities, defining the boundaries and data flows of the platform.

Overview of Entities and Data Flows

1. Artisan (External Entity)

The Artisan is the primary provider of goods on the platform. The data flows between the Artisan and the System include:

- **Inbound to System:** The Artisan provides registration details, manages product listings (creation/editing), initiates new auctions, and submits order status updates (e.g., changing status to "shipped").
- **Outbound from System:** The system provides the Artisan with essential information, including product details, updates on active orders, auction alerts (such as new bids), and system confirmations.

2. Customer (External Entity)

The Customer is the end-user purchasing goods. Their interactions are focused on the storefront:

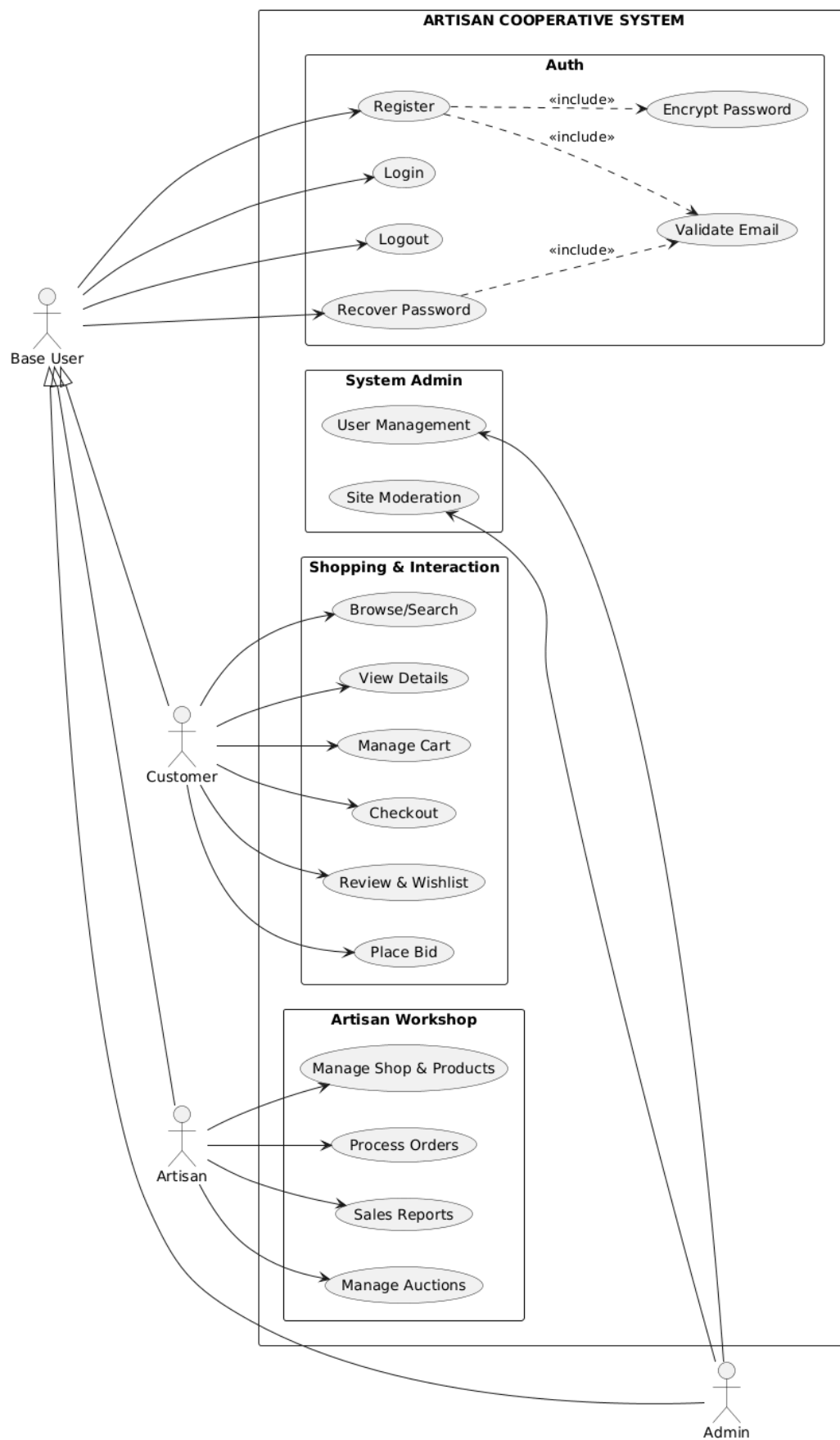
- **Inbound to System:** The Customer submits data for registration and login, executes product searches, performs cart actions (add/remove), completes the checkout process, places bids on auctions, and leaves reviews or wishlist updates.
- **Outbound from System:** The "Receive Info" flow represents the system delivering catalog data, order status tracking, bid results, and confirmation emails back to the user.

3. Admin (External Entity)

The Admin represents the governing body of the cooperative, ensuring smooth operations:

- **Inbound to System:** The Admin sends commands to manage the overall platform, including user accounts, product moderation, order dispute resolution, and auction oversight.
- **Outbound from System:** The system provides the Admin with comprehensive sales reports, system logs, and notification alerts regarding platform performance and activity.

Use Case Diagram



This image is a **Use Case Diagram** for the **Artisan Cooperative System**. It provides a high-level view of the system's functional requirements by illustrating the interactions between different types of users (**Actors**) and the specific tasks they perform (**Use Cases**).

Structure and Organization

1. Actors and Hierarchy

The diagram employs an inheritance model for the users:

- **Base User:** A general actor representing shared functionality.
- **Customer & Artisan:** These actors inherit from the "Base User," meaning they can perform all the core authentication tasks in addition to their specific roles.
- **Admin:** A specialized actor responsible for governance and oversight.

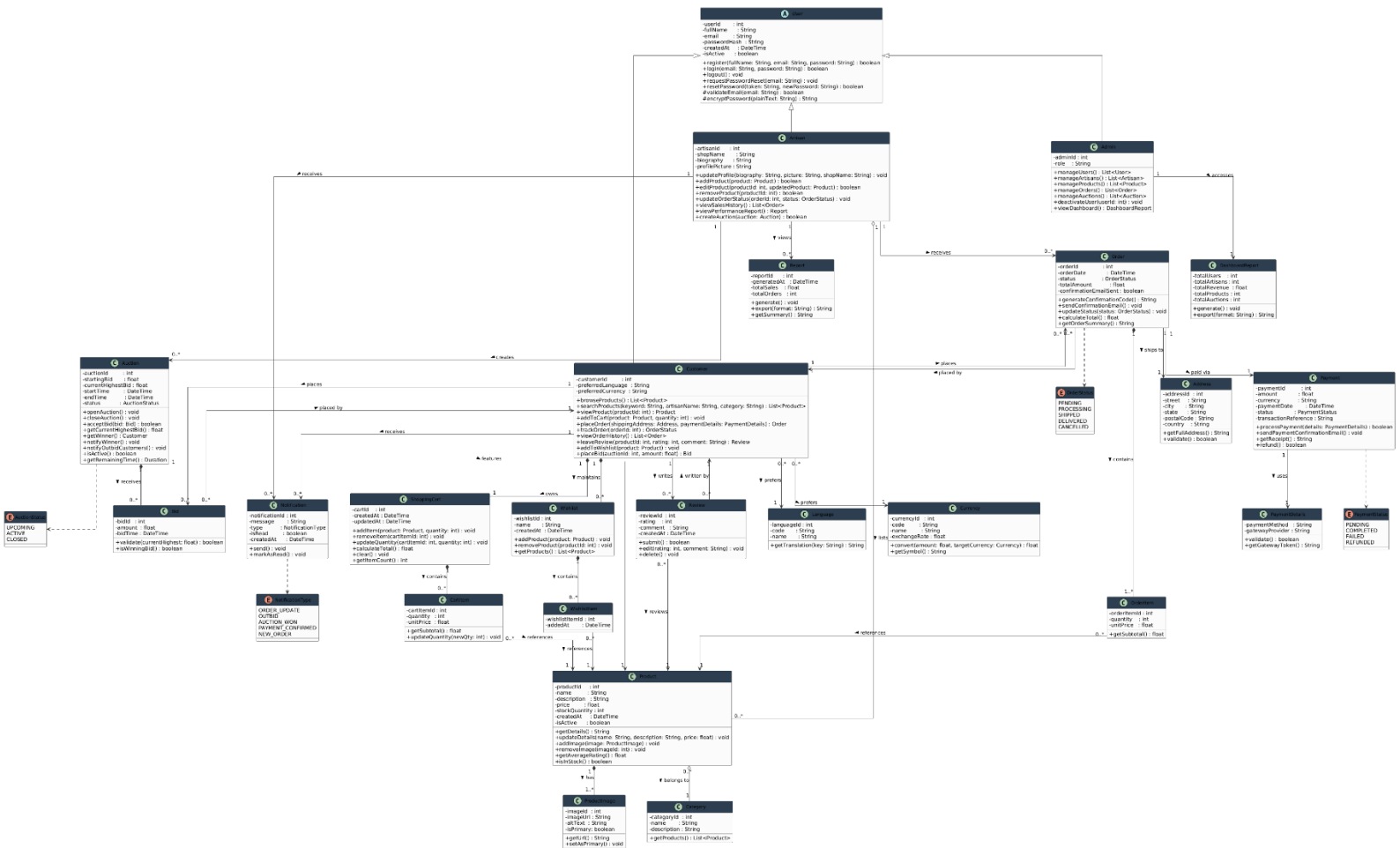
2. System Boundaries and Subsystems

The system is divided into four logical modules (subsystems) to organize functionality:

- **Auth (Authentication):**
 - Covers essential entry/exit functions: **Register, Login, Logout, and Recover Password.**
 - **Relationships:** Includes secondary processes like **Encrypt Password** and **Validate Email**, which are mandatory (<<include>>) steps during registration or recovery.
- **Shopping & Interaction (Customer Role):**
 - Focuses on the buyer's journey: **Browse/Search, View Details, Manage Cart, and Checkout.**
 - Includes engagement features such as **Review & Wishlist** and **Place Bid** (for the auction system).
- **Artisan Workshop (Artisan Role):**
 - Focuses on the seller's business operations: **Manage Shop & Products, Process Orders, and Manage Auctions.**
 - Provides business intelligence through **Sales Reports.**
- **System Admin (Administrator Role):**
 - Focuses on platform maintenance and safety: **User Management** and **Site Moderation.**

- The Admin also maintains overarching access to the Base User functions to manage accounts.

Class Diagram



a **Class Diagram** for a **Cooperative Online Marketplace System**. It serves as a technical blueprint for the software's architecture, illustrating the objects, their data (attributes), their behaviors (methods), and how they interact with one another.

Here is a breakdown of the diagram's core components:

1. The User Hierarchy (Inheritance)

At the top level is an **Abstract Class User**, which contains shared properties like `userId`, `email`, and `passwordHash`. This class acts as a template for three specific roles:

- **Customer:** Focused on the shopping experience, with methods for browsing, searching, placing orders, managing a wishlist, and placing bids.
 - **Artisan:** The seller role, responsible for managing a shop profile, listing products, tracking sales, and creating auctions.
 - **Admin:** A governance role with elevated permissions to manage users, moderate products, and oversee the entire platform's health via a DashboardReport.
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2. Product and Catalog Management

The system centers around the **Product** class.

- **Composition:** A product "owns" its **ProductImages**; if a product is deleted, its images are removed from the system.
 - **Categorization:** Products are grouped into a **Category** (e.g., Pottery, Textiles) via an aggregation relationship, allowing for organized browsing.
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3. The Sales and Transaction Lifecycle

The diagram maps out a clear path from shopping to fulfillment:

- **Shopping Cart:** Each Customer owns one ShoppingCart, which contains multiple CartItems. These items reference the Product but exist independently within the cart.
 - **Orders:** Once a checkout is finalized, an **Order** is generated. It links a Customer to a specific **Address** and a **Payment**. The Order contains **OrderItems**, which are snapshots of the products purchased.
 - **Payments:** The system tracks payment through a **Payment** class, which uses **PaymentDetails**. Notably, the code includes a comment ensuring no sensitive card data is stored on the platform, reflecting the security requirements.
-

4. Auction System

The diagram includes a specialized **Auction** module:

- **Auction** classes manage the lifecycle of a bidding event (UPCOMING, ACTIVE, CLOSED).
- An auction receives multiple **Bids**, which are placed by Customers.
- The logic includes methods for accepting bids, validating that new bids are higher than the current one, and notifying winners.

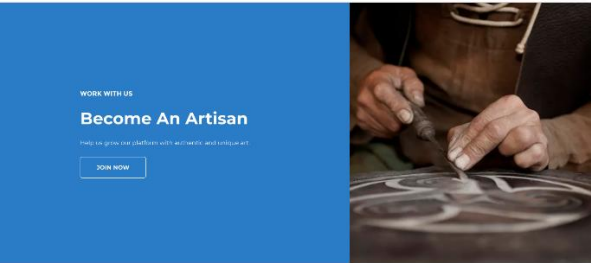
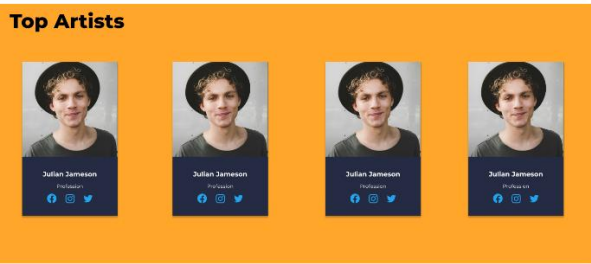
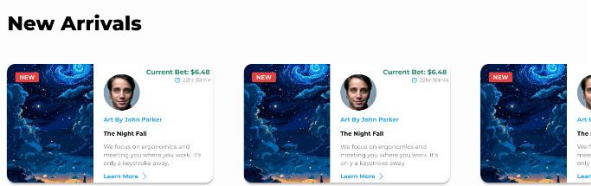
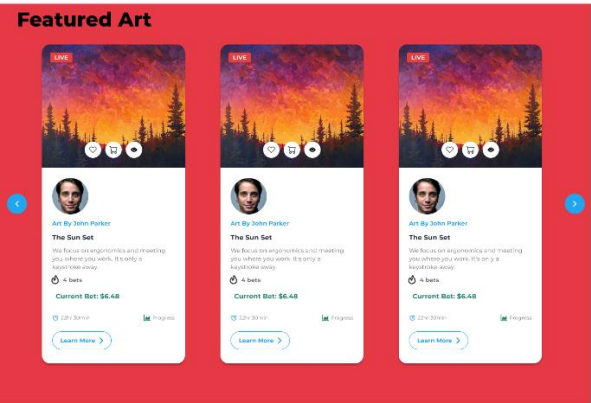
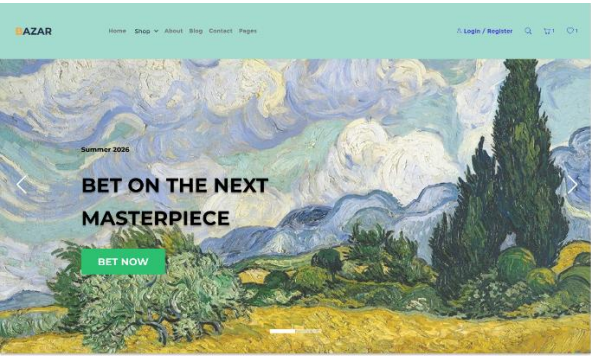
5. Engagement and Localization

- **Reviews:** Customers can write a Review for a specific Product, creating a many-to-one relationship between reviews and products.
- **Notifications:** The system uses a **Notification** class to alert users about order updates, outbid alerts, or payment confirmations.
- **Localization:** To support global reach, the system includes **Language** and **Currency** classes, allowing for dynamic translation and price conversion based on exchange rates.

6. Relationship Summary

- **Composition (Solid Diamonds):** Indicates strong ownership (e.g., an Order "owns" its OrderItems).
- **Aggregation (Hollow Diamonds):** Indicates a shared relationship (e.g., an Artisan "has" Products, but the product could theoretically exist or be archived separately).
- **Associations (Arrows):** Defines the flow of data, such as a Customer "placing" a Bid or an Order "shipping to" an Address.

Interface Design



UI Design Analysis: Cooperative Online Marketplace

The provided interface showcases a high-fidelity landing page for an artisan-focused digital marketplace. The design employs a modular, section-based layout that uses distinct color blocking to guide the user through different functional zones of the platform.

1. Visual Hierarchy and Color Strategy

The design utilizes a bold, high-contrast palette to define different content areas:

- **Hero and Header:** A clean, transparent navigation bar sits atop a large-scale artistic banner. The primary call-to-action (CTA) is a high-visibility green button labeled "BET NOW," emphasizing the auction aspect of the platform.
- **Featured Art Section:** Set against a deep red background, this area uses white product cards to create high legibility.
- **Artist Directory:** A warm golden-orange background differentiates the "Top Artists" section from product listings, shifting the focus from items to individuals.
- **Recruitment Zone:** A cerulean blue block serves as a dedicated portal for new artisans, utilizing a split-screen layout that balances text with high-detail photography of a craftsperson.

2. Component Design and Functional Elements

The interface relies heavily on standardized "Card" components to present complex data concisely:

- **Auction Cards:** Each card features a "LIVE" status indicator, artisan attribution (avatar and name), and real-time bidding data (number of bids, current price, and a countdown timer). Interactive icons for "Like," "Cart," and "Quick View" are grouped at the top of the image area.
- **Artisan Profile Cards:** These emphasize the human element of the cooperative. They include professional portraiture, the artisan's name, their specialty, and integrated social media links for further engagement.
- **Navigation and Utilities:** The header contains standard e-commerce utilities, including a search bar, shopping cart with a count indicator, and user authentication links (Login/Register).

3. User Experience (UX) Considerations

- **Clarity of Purpose:** By placing auction-specific data (current bid and time remaining) directly on the product cards, the design reduces the cognitive load on the user, allowing them to assess multiple auctions without leaving the main page.

- **Responsive Architecture:** The modular card-based system is inherently designed for responsiveness, suggesting a seamless transition to mobile views where cards would stack vertically.
- **Conversion Optimization:** Clear, bordered buttons like "Learn More" and "Join Now" provide unmistakable pathways for both customers and potential artisans to move deeper into the system.

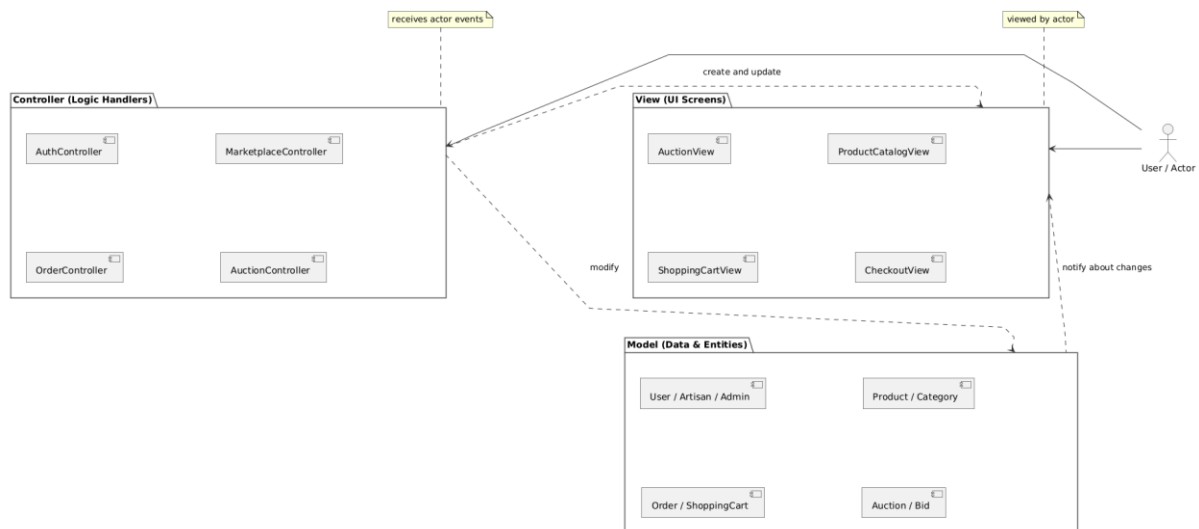
4. Information Architecture

The page follows a logical "AIDA" (Attention, Interest, Desire, Action) marketing funnel:

1. **Attention:** Large-scale art in the Hero section.
2. **Interest:** Featured live auctions and new arrivals.
3. **Desire:** Building trust through the "Top Artists" profiles.
4. **Action:** The "Become an Artisan" recruitment section at the bottom.

The interface successfully merges a commercial auction environment with a community-driven cooperative feel. The use of vibrant color blocking prevents the long-scrolling page from feeling repetitive, while the consistent card typography ensures that technical data—such as bid amounts and timers—remains the focal point of the marketplace experience.

Architecture of The System using MVC



This diagram illustrates the **MVC (Model-View-Controller) Architectural Pattern** tailored for the Cooperative Marketplace system. It defines the high-level organization of the platform into three specialized layers to ensure a clean separation of concerns:

1. View Layer (UI Screens)

This layer represents the **User Interface**. It contains specific screens such as the AuctionView, ProductCatalogView, and ShoppingCartView. It is responsible for presenting data to the user and is noted as being "**viewed by the actor.**"

2. Controller Layer (Logic Handlers)

Acting as the system's "brain," this layer handles user input and orchestrates the flow of the application. It includes specialized controllers like the AuctionController and AuthController. As indicated by the "**receives actor events**" note, it intercepts actions from the user (such as placing a bid) and determines the appropriate response.

3. Model Layer (Data & Entities)

This layer manages the **business logic and data** of the marketplace. It groups related entities from the class diagram—such as User/Artisan, Product/Category, and Auction/Bid. It maintains the state of the system and ensures data integrity.

Core Interaction Flow:

- **Action:** The **User/Actor** triggers events that the **Controller** receives.
- **Logic:** The **Controller** sends a command to **modify** the **Model** (e.g., saving a new bid).
- **Update:** The **Controller** coordinates the UI by **creating and updating** the **View**.
- **Feedback:** The **Model** ensures the UI remains accurate by **notifying the View** about state changes (e.g., updating the highest bid in real-time).

Conclusion

This project presented the design and analysis of an **E-Commerce Website for a Local Artisan Cooperative**, aiming to create a secure, scalable, and user-centric digital marketplace that supports both fixed-price sales and real-time auctions. By integrating e-commerce functionalities with auction mechanisms, the proposed system demonstrates how modern digital platforms can empower artisan communities to expand their market reach, increase engagement with customers, and operate more efficiently in a competitive online environment.

Through the application of a **DevOps-based Software Development Life Cycle (SDLC)**, the report highlighted the importance of continuous integration, continuous deployment, and continuous feedback in delivering reliable and high-quality software. Security and compliance were emphasized throughout all phases of development, ensuring the protection of sensitive user data, safe payment processing, and adherence to relevant regulatory requirements. Detailed functional, non-functional, business, and security requirements were identified and mapped to system components, ensuring that stakeholder needs were addressed in a structured and traceable manner.

The system design was supported by comprehensive models and diagrams, including context diagrams, use case diagrams, class diagrams, and an MVC architectural model. These artifacts provided a clear blueprint for system implementation and demonstrated how responsibilities are effectively separated to enhance maintainability, scalability, and performance. Additional features such as multilingual support, multi-currency handling, performance dashboards, and real-time auction updates further enhance user experience and operational efficiency.

In conclusion, this project illustrates the practical application of software engineering principles to a real-world problem, delivering a robust and extensible solution for cooperative-based online commerce. The proposed platform not only meets technical and business requirements but also promotes trust, transparency, and long-term sustainability for local artisans. Future enhancements may include advanced analytics and mobile application support further strengthening the platform's ability to adapt and grow in an evolving digital marketplace.